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Comparing hut-shaped-east-west array for fixed photovoltaic panels against conventional equator facing parallel rows for power output per unit field area

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ABSTRACT

Conventional method of placing fixed position photovoltaic panels is to use parallel rows facing the equator with tilt equaling latitude. In the current study, such layout is compared against an alternative hut shaped layout having east-west orientation. It is demonstrated that the alternative layout provides substantial advantage in terms of annual irradiation per land area. Equator facing parallel rows perform better only when the comparison is made for unit panel area. Moreover, the relative advantage of the proposed layout increases with latitude as the location is chosen away from the equator. Another aspect of conventional design that is challenged in the present study is the choice of optimal tilt angle. It is shown that the optimal tilt for a single equator facing panel is indeed close to the latitude angle, while the same is not true for an array where one row can shade the next. Here, a validated computer program is used to estimate the performance of different layouts of solar field by inputting irradiation availability data from standard National Solar Radiation Database. The study clearly shows the need to choose better layout for fixed position solar arrays which are being employed in large scale throughout the world.

Introduction

One of the major global grand challenges facing humanity is climate change. Use of the renewable sources of energy, such as solar energy, is indispensable for tackling the same. The worldwide solar energy generation is on an upward trend with an installed capacity of 714 GW in 2020 (IRENA, 2021). Despite the increasing trend of solar power installations, one impediment to expansion of photovoltaic (PV) power generation capacity is the requirement of large land area. The requirement of land increases linearly with installed capacity of a solar PV field. As per the Ministry of New and Renewable Energy (MNRE), Government of India, approximately 4–5 acres of land per MW of capacity is required to set up solar photovoltaic power generation even for high DNI locations (Ministry of New and Renewable Energy, 2021). The average total land requirement of large solar power plants in the United States is 7.5 acres per MW (Ong et al., 2013). For example, the Pavagada Solar Park, with a capacity of 2 GW in Karnataka, India, is spread over a total area of 13,000 acres (Solar Park, n.d.).

Rapid growth of solar PV power generation was made possible due to decreasing cost of the PV panels (IRENA, 2019; Kavlak et al., 2018). Nonetheless, larger capacity PV fields require larger land area, the cost of which keeps on increasing (Anna & Arts, 2019; Sampathkumar et al., 2015). A decade ago, the major portion of the capital investment required for a solar PV field was due to the cost of the PV panels (Hearps & McConnell, 2011). Presently, due to the aforementioned land cost hike and decreasing cost of the PV panels, the cost of land is a critical component which determines the feasibility of a PV installation. The reversal of the relative costs of panels and land has reversed the priority of the engineering optimization. Moreover, in the future, the growing land area usage by solar installations will cause an indirect impact on the emission associated with a PV installation (van de Ven et al., 2021). Therefore, the reduction in land area usage or extraction of maximum radiation from a given piece of land is imperative to reduce negative environmental impact. The arrangement of panels used to be dictated by the condition which maximized the annual irradiation per unit panel area. However the altered priority now requires to maximize the irradiation per unit land area. Alternatively, irradiation per land area can be maximized by optimizing the number of panels that can be fit in a given piece of land.

One of the major focus in the literature has been maximization of solar irradiation on a fixed angle non-sun-tracking solar panel. In these studies, the primary objective was to find the optimum tilt angles for arranging solar panels to obtain maximum irradiation (Bailek et al.,

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2018; Benghanem, 2011; Le Roux, 2016; Yadav & Malik, 2015). The state of the art of fixed solar panel layout is to orient the panels toward the equator (Equator Facing abbreviated as EF) at a tilt angle equal or close to the latitude for maximum irradiation annually (Cigala, 2016; Stapleton & Neill, 2012; *Handbook for rooftop solar development in Asia*, 2019). For the northern hemisphere, the orientation is southward. The parallel rows of panels are used for large-scale solar energy harvesting units where the same logic as used for a single panel is extended for designing the layout. However, the layout for maximum energy collection by a single panel does not necessarily convert to maximum yield by a field due to inter-panel shading. With increasing land cost, optimizing the solar panel array to avail maximum irradiation from a given area of land is also essential. The number of rows of solar panels that can be fit in a given piece of land can be increased by considering alternative layouting concepts. Kafka and Miller proposed the dual angle solar harvest (DASH) method to optimize incident solar energy in conjunction with land use (Kafka & Miller, 2020). It is an alternative method for organizing large solar panel arrays with rows of solar panels arranged at two different tilt angles instead of one. By tilting panels at two different angles, adding more solar panels in the given land area is possible.

Here we propose an alternative layout that is not orientated towards the equator (i.e., south-facing for the northern hemisphere) but orientated towards the east or west direction. The alternate rows are faced towards east and west, making the arrangement resembling hut-tops (East-West orientated Hut Shaped abbreviated as EWHS). The relative advantages of such layout over the conventional one are analyzed for a tropical location as well as a high-latitude location. The optimal tilt angles for maximizing the annual solar irradiations are obtained under both the constraints - per panel area and per land area.

The article first describes the methodology used to find the incident solar radiation on a single panel followed by extension of the same to obtain the solar irradiation on an equator facing (EF) array and east-west oriented hut-shaped (EWHS) array. Results are discussed based on different design criteria and locations, followed by specific recommendations for the layout designers of the fixed angle photovoltaic fields.

Methodology

Instantaneous irradiance data including direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), and global horizontal irradiance (GHI) for a given location with one hour interval are obtained from National Solar Radiation Database (NSRDB) from NREL, USA (NSRDB Data Viewer, n.d.). Hourly instantaneous irradiance data are considered as the average irradiance for a particular hour. The average irradiance data multiplied by time provides the irradiation available in that hour in units of MJ/m^2.

Estimation of annual irradiation on a single panel section explains the method used to find the amount of solar irradiation on a single south-faced (i.e., orientated towards the equator for panels located in the northern hemisphere) solar panel which, in turn, is used to determine the optimum tilt angle that maximizes the annual energy flux. Estimation of annual irradiation on an array of panels section discusses the methods associated with finding irradiation on an array of solar panels. One of the major hypotheses of the current study is that the optimal tilt angle for a single PV panel is not the same for an array of panels. The mathematical basis of the hypothesis is presented in Estimation of annual irradiation on a single panel and Estimation of annual irradiation on an array of panels sections. The second major hypothesis is the conventional orientation of facing the equator (i.e., south facing for the northern hemisphere and vice-versa) is not the optimal orientation when annual irradiation per unit land area is the metric of interest. To test this second hypothesis, two orientations are analyzed: (i) conventional equator-faced array (EF) and (ii) east-west faced hut-shaped (EWHS) array.

Estimation of annual irradiation on a single panel

State of the art algorithm for installing arrays of fixed (i.e., non-tracking) photovoltaic panels is to face the equator (i.e., in the northern hemisphere, panels are oriented toward the south and in the southern hemisphere, toward the north). Fig. 1 schematically shows the conventional orientation of a single solar panel in the northern hemisphere facing the south direction. Total solar irradiation on a panel constitutes of three components: direct, diffuse, and ground reflected irradiation. Total irradiation is obtained from Eq. (1) (Duffie et al., 1985).

$$I_{TOT} = I_{DIR} + I_{DIF} + I_g \tag{1}$$

where I_{DIR} is the direct irradiation, I_{DIF} is the diffuse irradiation, and I_g is the ground reflected irradiation.

From Hay (1993), direct irradiation on a tilted panel is obtained as shown below.

$$I_{DIR} = DNI \ \ cos\ \theta_i \tag{2}$$

where θ_i is the angle of incidence which is computed as follows (Duffie et al., 1985).

$$\begin{aligned}\cos\theta_i = {} & \sin\delta\ \sin\phi\ \cos\beta - \sin\delta\ \cos\phi\ \sin\beta\ \cos\gamma_c + \cos\delta\ \cos\phi\ \cos\beta\ \cos\omega \\ & + \cos\delta\ \sin\phi\ \sin\beta\ \cos\gamma_c\ \cos\omega + \cos\delta\ \sin\beta\ \sin\gamma_c\ \sin\omega\end{aligned}$$

Here, γ_c is the surface azimuth angle, δ is the declination angle, ϕ is the latitude, ω is the hour angle, and β is the tilt angle of the solar panel.

To obtain the second term of Eq. (1), the basic diffuse radiation model was proposed by Klucher (Jakhrani et al., 2013; Klucher, 1979; Pandey & Katiyar, 2011). Appelbaum et al. modified the Klucher model by introducing corrections in circumsolar brightening and horizon brightening (Appelbaum et al., 2019). Eq. (3) provides the state of the art diffuse irradiation estimator on a tilted panel as per Appelbaum model (Appelbaum et al., 2019).

$$\begin{aligned}I_{DIF} = {} & DHI \times F_{c\ -\ sky} \times \left[1 + F \times Y2 \times sin^3\left(\frac{\beta}{2}\right)\right] \\ & \times \left(1 + Y1 \times F \times cos^2\theta_i\ sin^3\theta_z\right)\end{aligned} \tag{3}$$

where $F = 1 - \left[\frac{DHI}{GHI}\right]^2$, which is a modulating factor, maximum under clear skies and minimum under overcast skies, F_{c-sky} is the view factor from the panel to the sky, Y1 is the circumsolar correction factor, Y2 is the horizon brightening correction factor and θ_z is the solar zenith angle. The panel-to-sky view factor for a single panel tilted at an angle β is given by, $F_{c-sky} = \frac{1+\cos\beta}{2}$ (Duffie et al., 1985).

The first correction factor in Eq. (3) (Y1) is related to the circumsolar diffuse radiation. When the sun is in the front of the panel, i.e., the DNI is non-zero, the value of Y1 is unity (Appelbaum et al., 2019). On the other hand, when the sun is behind the panel, the value of Y1 is zero. In case of

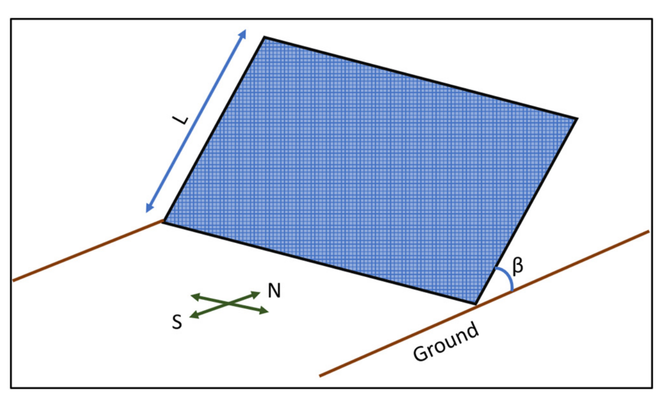

Fig. 1. Schematic diagram of a single solar panel facing south direction.

a single panel, nothing comes between the panel and the horizon as the ground is considered flat. Therefore the second correction factor (Y2), which relates to the horizon brightening, will be unity (Appelbaum et al., 2019).

The third term in Eq. (1), i.e., the ground reflected irradiation available to the solar panel, is estimated by Eq. (4) (Hay, 1993).

$$I_g = GHI \times F_{c-ground} \times \rho \tag{4}$$

where ρ is the ground surface albedo and $F_{c-ground}$ is the view factor from the panel to ground. In the current study, the value of the ground surface albedo is taken to be 0.2 (Hatfield et al., 2004). The panel-to-ground view factor is given by (Duffie et al., 1985).

$$F_{c-ground} = \frac{1 - cos\beta}{2}$$

Eqs. (1)–(4) are used to find the solar irradiation available at the chosen location for an hour. The hourly irradiation values are summed over an entire year to obtain the annual irradiation on a single fixed panel. The same exercise is repeated for various tilt angles ranging between 0° and 75°.

Estimation of annual irradiation on an array of panels

Photovoltaic power generation fields comprise of multiple rows of fixed solar panels. Whenever there is an array, as opposed to a single panel, irradiation on a particular panel is affected by the neighbouring panels. The major factor becomes the inter-panel shading particularly at low solar altitude angles in the early morning and late afternoon hours. To avoid this inter-panel shading, gaps are provided between the panels, the value of which is chosen based on the target working hours. When such gaps are introduced between the panels, the number of panel that can fit in a given field area reduces. Hence, it decreases the effective irradiation, which gets converted to photovoltaic power, per unit field area. Thus, to obtain a meaningful metric of comparison for arrays of PV panels, here we present the methodological modifications, in comparison with a single fixed panel (Estimation of annual irradiation on a single panel section), which are essential for estimating the effective irradiation per field area. As a model system for comparison, we consider 100 rows of solar panels in one array, with each row having a width, W, of 100 m. Height of each solar panel is taken as 2 m.

In hut shaped array, two rows of solar panels are arranged back to back to form a hut-shaped row. 100 panels are arranged in this way to make 50 hut-shaped rows. In the south-faced array model, all the 100 rows are kept parallel facing towards the south. The south-faced array model (EF) and east west oriented hut-shaped array model (EWHS) are shown in Fig. 2(a) and (b), respectively.

Placing panel rows adjacent to each other will affect the amount of solar irradiation available to the panels in two ways. Firstly, a row will cast a shadow on the row immediately behind, resulting in the reduction of direct irradiation. The shading will be present identically on all subsequent rows other than the first one because of the parallel placement of the rows. Secondly, the row in the immediate front will mask the view of the sky for all rows starting from the second one, thus reducing the diffuse irradiation available to them.

First, we will consider the inter-row shading issue. To ensure no shading of direct irradiation to the second and subsequent rows, adequate spacing needs to be provided between each pair of rows. For the south-faced array, inter-row spacing is taken as the distance between panels to be unshaded on the winter solstice between 10 am and 2 pm solar time, which ensures no other day of the year will see the inter-row shading for four hours around the solar noon (*Utility-Scale Solar Photovoltaic Power Plants: a project developer's guide*, 2015). For a hut-shaped array, inter-row spacing is the distance between the feet of two adjacent hut-shaped rows. Spacing is calculated such that there will not be any shadow cast by one hut-shaped row on another between 10 am and 2 pm solar time throughout the year. Fig. 3 shows the inter-row spacing, S_p, for south faced array and hut shaped array. In this figure, L_g is the length of a row projected along the ground and G is the free space gap between two rows. Inter-row spacing, S_p, is the sum of L_g and G.

The inter-row spacing or panel spacing for both models of solar panel arrays can be represented by Eq. (5) (Bany & Appelbaum, 1987).

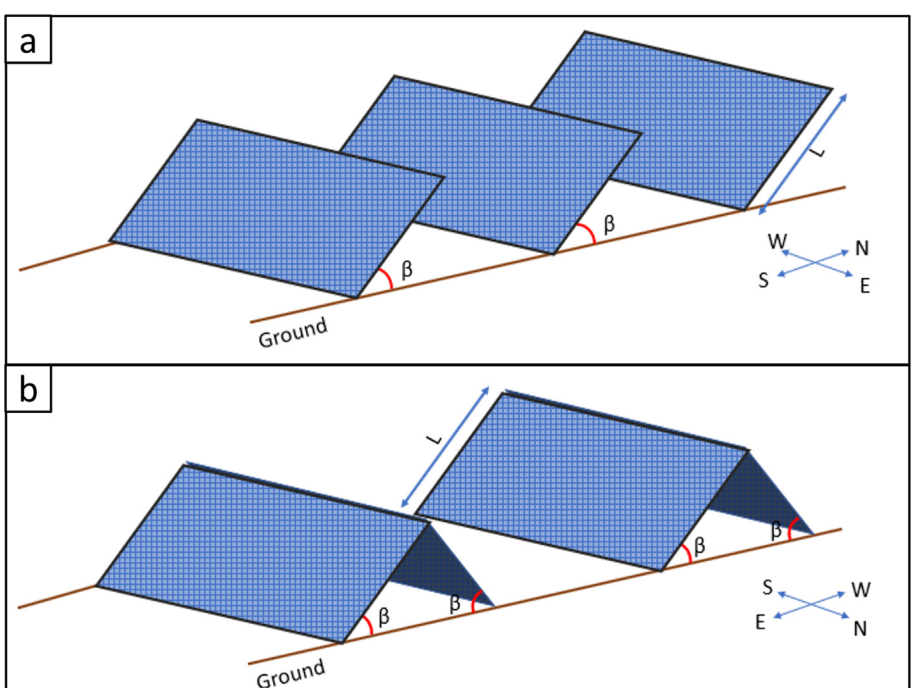

Fig. 2. Schematic arrangements of solar panel array: (a) Equator-faced array (EF). (b) Hut-shaped array in East-West orientation (EWHS).

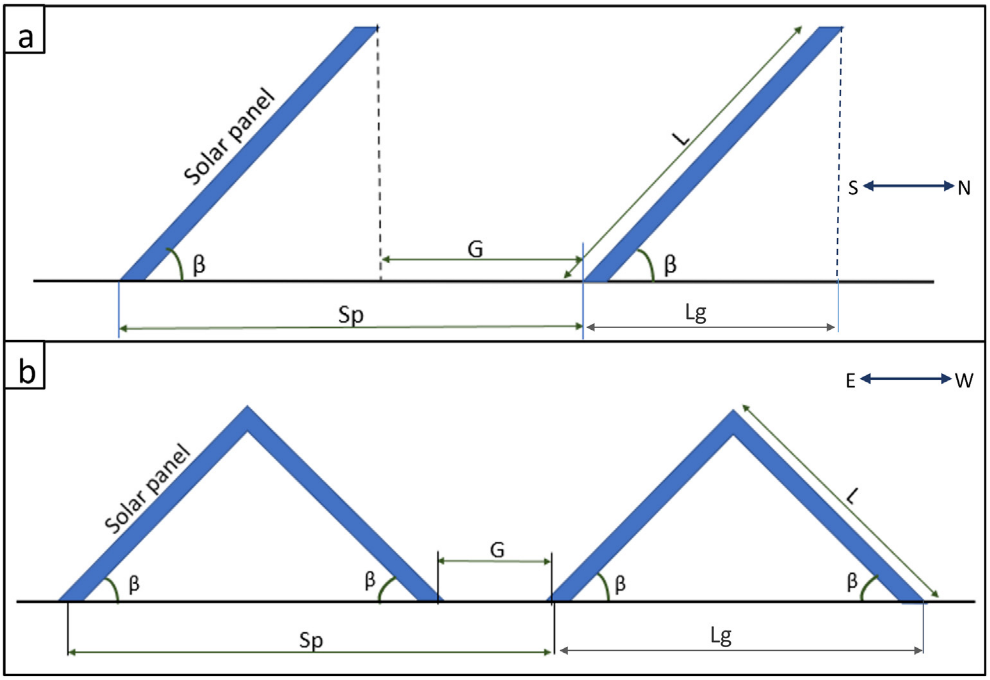

Fig. 3. Relevant geometrical parameters in an array layout: (a) South-faced array. (b) Hut-shaped array. Here, inter-row spacing is denoted by S_p; G is the gap between consecutive rows and L_g is the length of one row projected on the ground, which is assumed horizontal.

$$S_p = L \times \left[cos\beta + \frac{sin\beta \times \cos\left(\gamma_s - \gamma_c\right)}{tan\alpha} \right] \tag{5}$$

where β is the tilt angle, α is the solar altitude angle, L is the length of solar panel, γ_s is the solar azimuth angle, and γ_c is the surface azimuth angle.

Even after keeping a space of length S_p between adjacent rows, there would still be shading in the morning and evening (beyond the window of 10 am and 2 pm solar time). The length and width of this shadow cast on a row of solar panels can be represented by Eq. (6) and Eq. (7), respectively (Bany & Appelbaum, 1987).

$$L_{sh} = L - \left(\frac{S_p \times tan\alpha}{cos\beta\ tan\alpha + sin\beta\ \cos\left(\gamma_s - \gamma_c\right)} \right) \tag{6}$$

$$W_{sh} = W - \left(\frac{S_p \times sin\beta \mid \sin\left(\gamma_s - \gamma_c\right) \mid}{cos\beta\ tan\alpha + sin\beta\ \cos\left(\gamma_s - \gamma_c\right)} \right) \tag{7}$$

For the area shaded in a row, $L_{sh} \times W_{sh}$, direct irradiation will be zero. For the area unshaded, direct irradiation will be the same as in Eq. (2).

The first row in the south-faced solar array and end sides of the hut-shaped solar array are free from inter-row shading.

Next, we consider the issue of alteration of view factor between the panel and the sky. The front row will mask the view of sky for subsequent rows behind it, thereby reducing the diffuse irradiation. The reduction in the diffuse irradiation depends on the sky view factor of each row which, in turn, depends on inter-row spacing, height of panel, and the layout of the array (i.e., EF vs. EWHS).

Estimation of view factors

The sky view factor and ground view factor for an equator facing (EF) array and east-west hut-shaped (EWHS) array are found using the cross string rule (Modest, 2013). First, we are considering EF solar array. A schematic diagram of EF solar array is shown in Fig. 4. In this figure, L1 is the distance between bottom of panel 1 and bottom of panel 2, L2 is the distance between top of panel 1 and top of panel 2, L3 is the distance between top of panel 1 and bottom of panel 2, L4 is the distance between bottom of panel 1 and top of panel 2, S_p is the panel spacing and β is the tilt angle. As the panels are parallel,

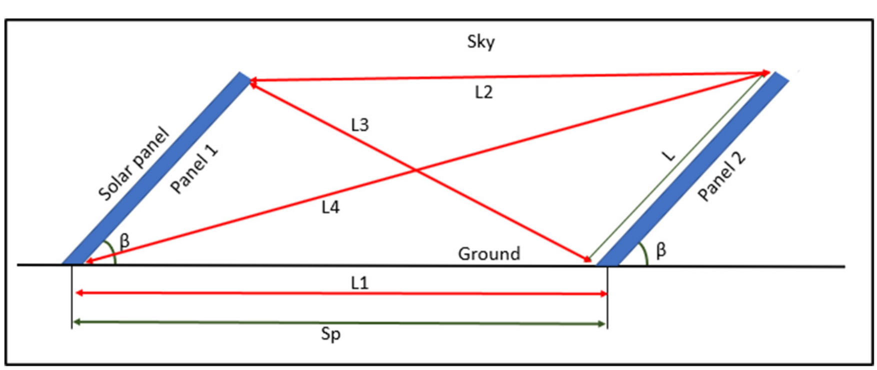

Fig. 4. Schematic diagram of a EF solar array. L1, L2, L3, and L4 are the distances between corners of two panels in adjacent row.

$$L1 = L2 = S_p$$

$$L3 = \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}$$

$$L4 = \sqrt{(Lsin\beta)^2 + (S_p + Lcos\beta)^2}$$

The view factor from panel 2 to panel 1 is represented by $F_{2\text{-}1}$.

$$F_{2-1} = \frac{(L3 + L4) - (L1 + L2)}{2 \times L}$$

The view factor from panel 2 to ground is denoted by $F_{2\text{-ground}}$ and is given by,

$$F_{2-ground} = \frac{(L + L1) - (L4)}{2 \times L}$$

$$F_{2-ground} = \frac{L + S_p - \sqrt{(Lsin\beta)^2 + (S_p + Lcos\beta)^2}}{2 \times L}$$

From the view factor algebra, the sum of the view factors from panel 2 to the whole space is 1. i.e., $F_{2-ground} + F_{2-1} + F_{2-sky} = 1$. Where $F_{2\text{-sky}}$ is the sky view factor of panel 2.

$$F_{2-sky} = 1 - \left(F_{2-1} + F_{2-ground}\right)$$

$$F_{2-sky} = 1 - \frac{(L3 + L4) - (L1 + L2) + (L + L1) - (L4)}{2 \times L}$$

$$F_{2-sky} = \frac{L + S_p - \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}}{2 \times L}$$

Since all the panels are placed at the same tilt angle, sky view factor and ground view factor of all the panels except the front row will be the same as that of panel 2. So in general ground view factor and sky view factor of a panel in an EF solar array is given by Eq. (8) and Eq. (9) respectively.

$$F_{c-ground} = \frac{L + S_p - \sqrt{(Lsin\beta)^2 + (S_p + Lcos\beta)^2}}{2 \times L} \tag{8}$$

$$F_{c-sky} = \frac{L + S_p - \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}}{2 \times L} \tag{9}$$

Next we consider EWHS array. A schematic diagram of EWHS array is shown in Fig. 5. Similar to the EF array, L1, L2, L3 and L4 are the distance between corners of panels in adjacent rows. S_p is the panel spacing and β is the tilt angle.

$$L1 = S_p - 2L\,cos\beta$$

$$L2 = S_p$$

$$L3 = L4 = \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}$$

The view factor from panel 1 to panel 2 is represented by $F_{1\text{-}2}$.

$$F_{1-2} = \frac{(L3 + L4) - (L1 + L2)}{2 \times L}$$

The view factor from panel 1 to ground is denoted by $F_{1\text{-ground}}$ and is given by,

$$F_{1-ground} = \frac{(L + L1) - (L3)}{2 \times L}$$

$$F_{1-ground} = \frac{L + S_p - 2L\,cos\beta - \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}}{2 \times L}$$

When we place hut-shaped rows without any gap between them, $S_p = 2L\,cos\beta$. The ground view factor of panel 1, $F_{1-ground} = 0$.

The sum of the view factors from panel 1 to the whole space is 1. i.e., $F_{1-ground} + F_{1-2} + F_{1-sky} = 1$. Where $F_{1\text{-sky}}$ is the sky view factor for panel 1.

$$F_{1-sky} = 1 - \left(F_{1-2} + F_{1-ground}\right)$$

$$F_{1-sky} = 1 - \frac{(L3 + L4) - (L1 + L2) + (L + L1) - (L3)}{2 \times L}$$

$$F_{1-sky} = \frac{L + S_p - \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}}{2 \times L}$$

When the hut-shaped rows are placed without any gap between them, $S_p = 2L\,cos\beta$. In that case, the sky view factor of panel 1 is the following.

$$F_{1-sky} = \cos\beta.$$

Since all the panels are placed at the same tilt angle, sky view factor and ground view factor of panel 2 will be the same as that of panel 1. Also, we note that the sky view factor for both the south-faced solar array (EF) and the hut-shaped solar array (EWHS) is the same which is represented by Eq. (9). The ground view factor of a panel in an EWHS array is given by the Eq. (10).

$$F_{c-ground} = \frac{L + S_p - 2L\,cos\beta - \sqrt{(Lsin\beta)^2 + (S_p - Lcos\beta)^2}}{2 \times L} \tag{10}$$

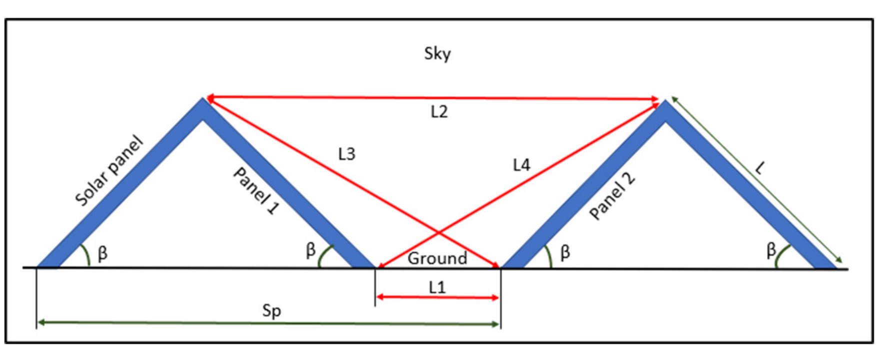

Fig. 5. Schematic diagram of an EWHS array. L1, L2, L3 and L4 are the distance between corners of two panels in adjacent row.

There will not be any sky view masking for the first row in the south-faced solar array and end sides of the hut-shaped solar array. For these rows, the entire sky dome will be visible, as in the case of a single panel. Therefore, the sky view factor found in Estimation of annual irradiation on a single panel section will be applicable for these panels.

The sky view factor discussed above can be used in Eq. (3) discussed in Estimation of annual irradiation on a single panel section to find the diffuse irradiation available to the arrays. The circumsolar factor and horizon brightening factor are row specific for an array. For the first row in the south-faced solar array and end sides of the hut-shaped solar array, the circumsolar factor and horizon brightening factor will be the same as that discussed for a single panel in Estimation of annual irradiation on a single panel section. For the remaining rows, at first, we consider the circumsolar factor. The front row will block the circumsolar irradiation available to the rows behind, as shown in Fig. 6. In this figure, α is the solar altitude angle, and ψ is the obscuring angle from the sun on the second row caused by the front row. If the altitude angle, α, is less than the obscuring angle ψ, the back row is blocked from circumsolar irradiation. If α is greater than ψ, circumsolar radiation will reach the back row. Circumsolar factor can be represented by Eq. (11) (Appelbaum et al., 2019).

$$Y1 = \begin{cases} 1 & \text{if } \psi < \alpha \\ 0 & \text{if } \psi \geq \alpha \end{cases} \tag{11}$$

where,

$$\psi = tan^{-1}\left(\frac{L\, sin\beta}{S_p - L\, cos\beta}\right) \tag{12}$$

For the second and subsequent rows, the horizon brightening effect are blocked by the front row. Hence, the horizon brightening factor for these rows is zero.

Next, the ground reflected irradiation available to these arrays is considered. Eq. (4) can be used to find the ground reflected irradiation for both models (i.e. EF and EWFS) of solar arrays. The ground view factor, $F_{c-ground}$, is both layout-specific and row-specific. For the first row in the south-faced solar array and end sides of the hut-shaped solar array, $F_{c-ground}$ is the same as that for a single solar panel. For the remaining rows, the ground view is given by Eq. (8) and Eq. (10).

Total irradiation can be found by multiplying each term in Eq. (1) by the panel area. In the case of direct irradiation, the area shaded by inter-row shading is subtracted from the total area. Dividing total irradiation on the array by the land area used results in the irradiation per land area.

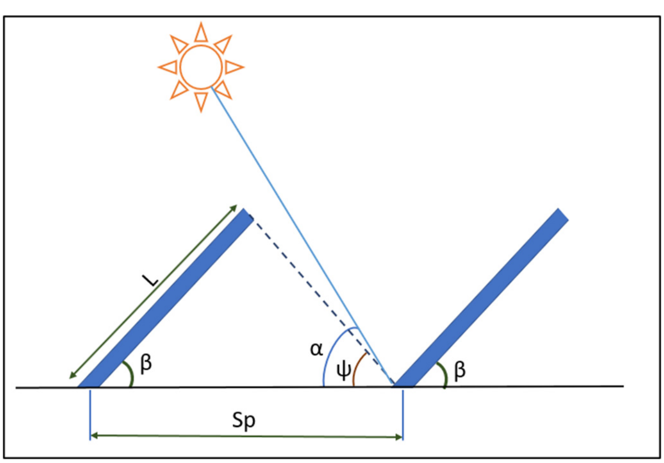

Fig. 6. Schematic diagram demonstrating blocking of direct irradiation and circumsolar irradiation. The first row will block the direct irradiation and circumsolar irradiation on the second row when $\psi > \alpha$. If $\psi < \alpha$, direct as well the circumsolar diffuse irradiation is available to the entire second row.

Estimation of land area

Fig. 7 shows the schematic diagram of an equator facing (EF) solar array (Fig. 7(a)) and an east-west hut-shaped (EWHS) solar array (Fig. 7(b)). In this figure, S_p is the panel spacing, β is the tilt angle, L_{TOT} is the total length of array and L_g is the projected length of a single row on the ground.

From Fig. 7, it is clear that all the rows except the last row will take a length of S_p along the ground and the last row uses L_g length. The total length of the array will be the following.

$$L_{TOT} = [S_p \times (\text{Number of rows} - 1)] + L_g$$

For an EF array, $L_g = L\, cos\beta$ and for an EWHS array, $L_g = 2L\, cos\beta$. Therefore the land area used by EF array is the following.

$$\text{Land area used by EF array} = \{[S_p \times (\text{Number of rows} - 1)] + (L\, cos\beta)\} \times W \tag{13}$$

The land area used by an EWHS array is as follows.

$$\text{Land area used by EWHS array} = \left\{[S_p \times (\text{Number of rows} - 1)] + (2 \times L\, cos\beta)\right\} \times W \tag{14}$$

The total solar irradiation on the array and solar irradiation per land area are calculated for every hour throughout the year, and the hourly values are summed to get the yearly total irradiation.

Method validation

Previous studies in hut-shaped solar array design are not available to be recomputed. Therefore we are benchmarking our method by comparing it with the results obtained in a study done on a south-faced solar panel at Abu Dhabi by Jafarkazemi and Saadabadi (Jafarkazemi & Saadabadi, 2013). For comparison purposes, we are taking yearly and monthly optimum tilt angles and the irradiance per panel area at these tilts. In Ref. (Jafarkazemi & Saadabadi, 2013), tilt angles and irradiance were calculated using Erbs model, KT model, and monthly average daily total radiation data provided by NASA Surface Meteorology and Solar-Energy model. In current study, optimum tilt angles and irradiance are calculated using the methods discussed in Estimation of annual irradiation on a single panel section. We used hourly radiation data from NREL's National Solar Radiation Database (NSRDB) as the input and Klucher radiation model for prediction (NSRDB Data Viewer, n.d.; Klucher, 1979). The optimum tilt angles and irradiance data are also available from the National Aeronautics and Space Administration (NASA) Langley Research Center (LaRC) Prediction of Worldwide Energy Resource (POWER) Project funded through the NASA Earth Science/Applied Science Program (POWER, n.d.).

Fig. 8 shows the comparison between the results of Ref. (Jafarkazemi & Saadabadi, 2013), data available from NASA LaRC POWER project, and the values found in the current study. Fig. 8(a) depicts the comparison of monthly and yearly optimum tilt angles. Yearly optimum tilt angle found in Ref. (Jafarkazemi & Saadabadi, 2013) is 22° which is very close to the optimum tilt found in current study (=21°) and NASA LaRC POWER data (=21°). Also this figure shows that all three studies are showing similar trends in monthly optimum tilt angles, where the minimum tilt angle is required for June and maximum for December.

Fig. 8(b) represents comparison of irradiance per panel area when the panel is tilted at the optimum tilt angles shown in Fig. 8(a). Irradiance per panel area found in current study for yearly optimum tilt angle is 6.2 KW-hr/m^2/day, while it is 5.9 kW-h/m^2/day from Jafarkazemi and Saadabadi (2013), and from NASA LaRC POWER data,

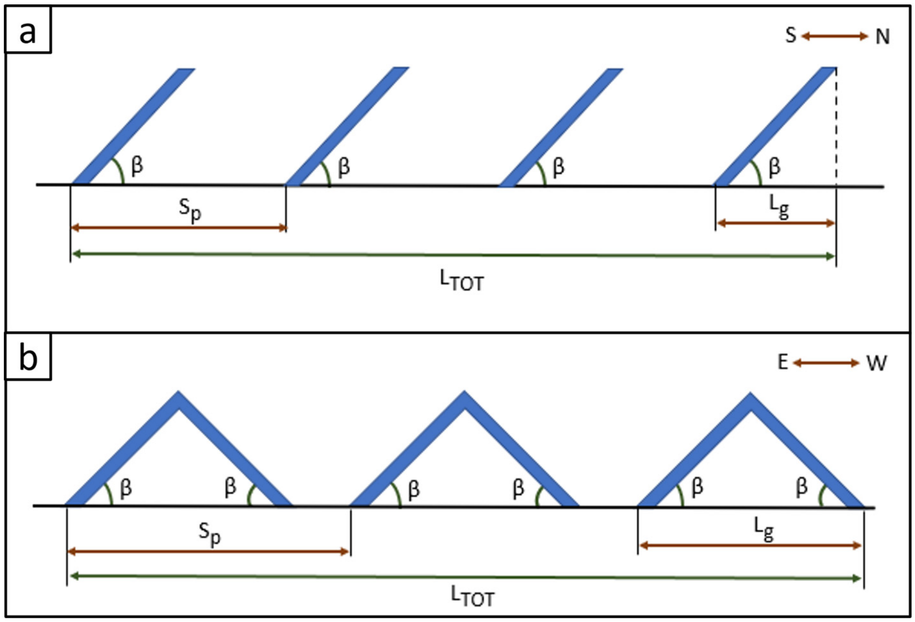

Fig. 7. Schematic diagram of solar arrays: (a) EF array. (b) EWHS array. Here L_{TOT} is the total length of array along ground, S_p is the panel spacing and L_g is the projected length of a single row on the ground.

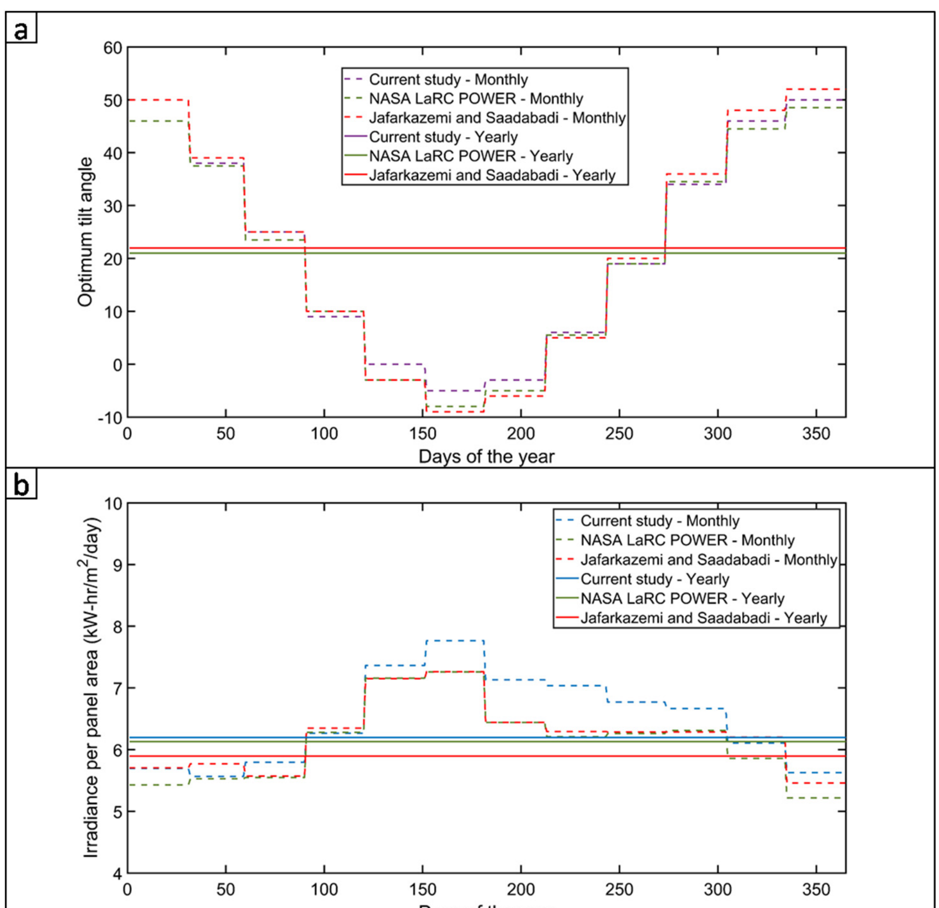

Fig. 8. Comparison of NASA SSE model (The data was obtained from the National Aeronautics and Space Administration (NASA) Langley Research Center (LaRC) Prediction of Worldwide Energy Resource (POWER) Project funded through the NASA Earth Science/Applied Science Program.), the study done by Jafarkazemi and Saadabadi, and the current study: (a) Optimum tilt angle vs. days of a year where monthly and yearly optimum tilt angles are compared (b) Irradiance per panel area vs. days of a year, when the panels are oriented at optimum monthly and yearly tilt angles.

it is 6.1 kW-h/m^2/day. The dashed curves represent irradiance for monthly optimum tilt angles, which show the same trend for all three sources. The small variations in values between curves in both Fig. 8(a) and Fig. 8(b) is mainly due to the difference in models used and the input dataset.

Both the irradiance and the optimum tilt angles found in the current study, the study done by Jafarkazemi and Saadabadi, and data available from NASA LaRC POWER project are close. Therefore our methodology stands validated for calculating the irradiation available on panel surfaces.

Results and discussion

In this section, the comparison between different layouts is reported in terms of total annual solar irradiation received. One layout may outperform the other at a given time. Nonetheless, the relative preference gets altered depending on the season of a year and the hour of a day. Annual irradiation is independent of these temporal variations. It provides the relative comparison between different layouts in absolute term hence chosen as the performance metric in the present study. Another important point to note here is that the comparison is only meaningful when normalized by the total area. If a layout is replicated twice, the total irradiation should not be counted twice. Hence, in this section, all the comparisons are made in terms of annual total irradiation per area (MJ/m^2).

Optimum orientation for hut shaped array

It is essential to find the best orientation for a hut-shaped solar array for the purpose of comparing it with an equator facing array. Here we are analyzing irradiation per panel area and irradiation per land area on a 100 m wide, 50 row hut-shaped solar array arranged in four different orientations, East-West, North-South, North East-South West, and South East-North West. The arrays are arranged in such a way that there is no fixed minimum gap between adjacent rows. The irradiation is calculated using the methods discussed in Estimation of annual irradiation on an array of panels section.

The curves showing the comparison of these four directions are represented by Fig. 9. Fig. 9(a) compares the annual total irradiation per panel area available to the arrays. For all the hut-shaped solar panel arrays, irradiation per panel area is maximum when the tilt angle is 0° (= 6788.9 MJ/m^2), i.e., when the panels are placed horizontally parallel to the ground. Initially, for the lower tilt angles, the variation in irradiation between orientations is negligible. As the tilt angle increases, the North East-South West oriented hut-shaped solar array performs better than other orientations.

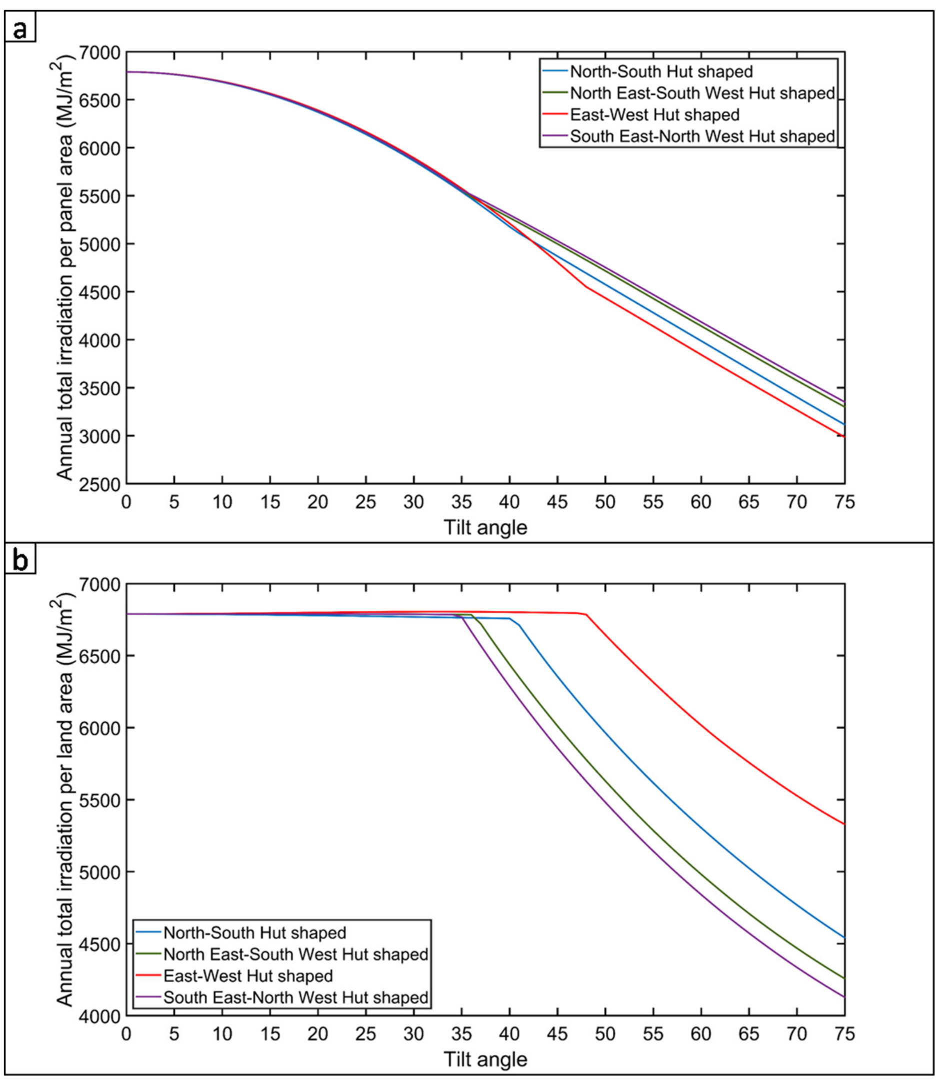

Fig. 9. Comparison of hut shaped solar arrays oriented in four different directions, North-South, North East-South West, and East-West and South East-North West: (a) Annual total irradiation per panel area v/s. tilt angle (b) Annual total irradiation per land area v/s. tilt angle.

Fig. 9(b) depicts the irradiation per land area as a function of tilt angle. The difference in irradiation between each orientation is negligible for smaller tilt angles up to 34° and grows with increasing tilt angle. This figure shows that the East-West faced hut-shaped (EWHS) array outperforms all the other orientations. The advantage of EWHS increases for higher tilt angles. The maximum irradiation per land area available for an east-west hut-shaped array is 6804.9 MJ/m^2 at 33° tilt angle.

The primary objective of this study is to optimize solar arrays for maximum annual yield with minimum usage of land area. Since East-West Hut Shaped (EWHS) solar array provides better output per land area (see Irradiation per unit land area section), we choose EWHS solar arrays to compare with the conventional equator facing arrays.

Irradiation on a single panel

At first, it is important to compare the conventional equator facing (EF) orientation for a single panel against the same for an array of multiple parallel panels. A single panel is used in many domestic rooftop installations, while an array of panels is for large scale power generation connected to the grid. State of the art practice is to follow the same tilt angle for both cases. In Fig. 10(a), it is shown that such a practice is not based on careful quantitative comparison. The blue line shows the variation of annual irradiation per panel area for an EF panel in an arbitrarily chosen location of Kolkata, India. With increase in tilt with respect to the horizontal plane, the annual irradiation value slowly increases first, reaches the maximum at 21° and then drops quickly for larger values of tilt. As the maximum is reached at 21° tilt, it is best to orient the panel towards the equator at that particular tilt to maximize the annual solar collection. Here we note that the optimal tilt is close to the latitude of the chosen location (Kolkata: 22.4°N, 88.4°E). The state of the art practice is to choose the optimal tilt angle to be equal to the latitude of the location for the equator facing orientation (Cigala, 2016; Stapleton & Neill, 2012; *Handbook for rooftop solar development in Asia*, 2019). Due to terrestrial variation of insolation and due to the oblate spheroid shape of the earth, we observe a slight variation in the optimal tilt.

On the other hand, the same is not true for an array of parallel panels. The red curve in Fig. 10(a) shows a similar trend (i.e., increasing first, decreasing subsequently) as the blue one. However, the maximum value is reached for different values of the tilt angle. For the red curve, the maximum corresponds to the tilt angle of 12°. Note that both the curves are normalized per unit panel area. Hence the effect of larger area for an array as compared to a single panel is ruled out as the reason behind the observed variation. In the case of an array, consecutive panels can shade each other during the hours of low solar altitude

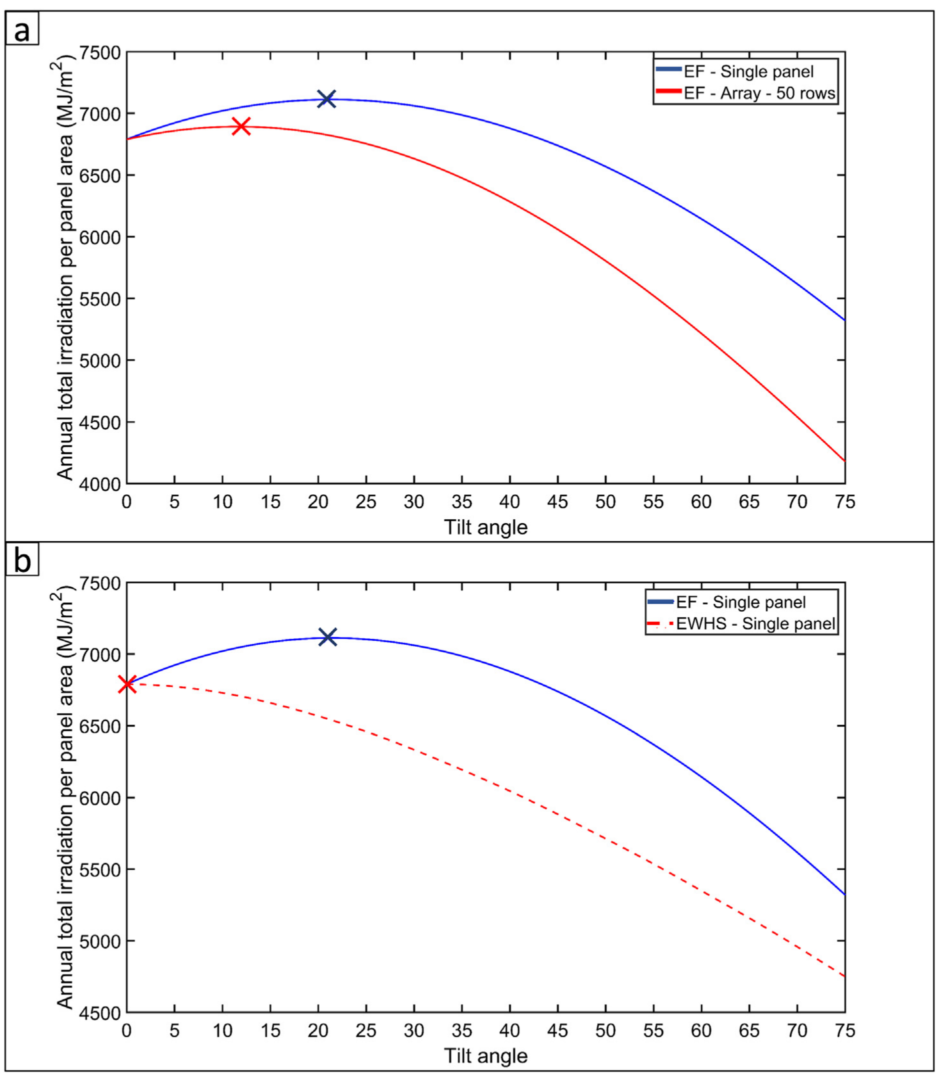

Fig. 10. Annual total irradiation per panel area: (a) Comparison between a single equator facing (south facing for Kolkata, India, a location in the northern hemisphere) solar panel and a south facing array (b) Comparison between a single equator facing solar panel and an east-west hut shaped panel (Place: Kolkata). The points corresponding to the maximum irradiation are shown with cross mark on each curve.

angles, which does not arise in the case of a single panel. Therefore, inter-panel shading causes the optimal tilt for an array to be different from that for a single panel. Thus, Fig. 10(a) shows clearly that the standard practice of tilting the equator faced arrays of solar panels by the same angle as a single panel is equivalent to underutilizing its potentiality substantially. For the location of Kolkata, an optimally orientated single EF panel provides 3.2 % more annual irradiation (7112 MJ/m^2) as compared to an optimally orientated array of EF panels (6892 MJ/m^2) when the absolute values are normalized by the panel area.

Next, we compare the annual performances of a single panel in the equator faced (EF) orientation with that in an East-West-Hut-Shaped (EWHS) orientation (see Fig. 10(b)). Again, the effect of larger area for a hut shaped layout is normalized by considering irradiation per unit panel area. In Fig. 10(b), the blue solid curve shows the performance variation of an EF (i.e., south facing for Kolkata) panel and the red dashed curve shows the same for an EWHS panel. Note that the blue curves for Fig. 10(a) and Fig. 10(b) are identical. For Fig. 10(b), it can be observed that the EWHS layout provides lower irradiation value for all tilt angles while the best performance is achieved for horizontal orientation (tilt = 0°), where the distinction between EWHS and EF is irrelevant. The optimally oriented EF panel provides 4.8 % higher annual irradiation (7112 MJ/m^2) in comparison with the EWHS panel (6789 MJ/m^2) for the location of Kolkata.

Irradiation per unit panel area

Use of single panel is only restricted for household applications. Any industrial-scale solar power generation plant consists of a large number of panels placed together in an open field. Whenever there are multiple rows of panels, inter-panel shading becomes relevant and the irradiation estimate is only accurate when such shading is enumerated and incorporated with precision.

The irradiation estimation can be normalized by two alternative methods – (a) per panel area and (b) per land area. These two metrics are not necessarily equivalent. Even for the same number of rows, which ensures the same panel area is employed, the area of land required will depend on the tilt of the individual rows as well as the gaps between consecutive rows. Hence, the aforementioned methods of normalization produce different comparison metrics.

In terms of picking the right metric for the layout optimization, the most important factor is the relative cost of solar panel vs. the price of land. When solar panels are very costly as compared to land, the design criterion should be to maximize annual irradiation per panel area so that for the same target of energy yield, minimum number of panels are used. However, for costly land and low-cost panels, the right choice of the objective function is to minimize the utilization of land for the same energy yield even when the number of panels is higher. The same is true when a fixed area of land, such as a large flat rooftop, is available on which the energy yield is to be maximized irrespective of the number of panels. Therefore, both the design criteria are important and need to be investigated. In the current study, we explore both of them. While the second criterion will be analyzed in Irradiation per unit land area section, here we look into the first criterion, i.e., irradiation per unit panel area.

Fig. 11(a) shows the values of the annual irradiation on EF and EWHS layouts for different number of rows and different tilt angles. Each solid line lies above its corresponding dashed line for all tilts, indicating that the EF layout is the correct choice for each of these cases. In terms of tilt angle, all the EWHS layouts show the maximal point when they are horizontal. It is to be noted that the difference between various EF layouts are minimal when the row numbers are beyond a few. In other words, the single row EF layout (equivalent to a single panel simulation shown in Fig. 10(a)) outperforms the same with two rows significantly; The two-row layout outperforms the five-row layout slightly less significantly; however, beyond the five-row layout, the increase in the number of rows hardly makes any quantitative difference in terms of the annual irradiation per panel area. The reason behind this trend is that the end effect becomes insignificant beyond a certain number of parallel rows and the inter-row effect dominates.

Moreover, the optimal tilt shown by the crosses (in the inset figure) on the curves corresponding to different EF-layouts show the following trend. The higher the number of rows, the lower the optimal tilt, while the change is minimal when the number of rows is large enough. The trend elucidates the fact that the inter-row shading dictates the performance of a layout rather than the end effect, where the lower tilt is preferred due to the lower total duration of shading.

While Fig. 11(a) shows the trend for a layout where the consecutive rows are not spaced with any minimum gap in their horizontal projections, Fig. 11(b) depicts the same where fixed gaps of 0.5 m, 1 m, 2 m, and 5 m are considered between consecutive rows. There are two justifications for keeping a certain distance (say, G as shown in Fig. 3) between two consecutive rows of solar panels. First one is the possibility of less shading as the shadow of the front row can lower the performance of the next row only when the shadow length is more than G. Secondly, the gap provides the clearance for the workmen or devices to access each individual panel for regular cleaning and maintenance.

In the inset figure of Fig. 11(b), the optimal tilts for arrays are marked by crosses on each EF curve shown in Fig. 11(b). The higher the gap, the higher the optimal tilt. Moreover, with increasing gap, the change in optimal tilt reduces. In other words, the difference in optimal tilt between array with 0.5 m gap and array with 1 m gap is higher than the difference in optimal tilt between array with 2 m gap and array with 5 m gap. This is because while increasing the gap from 0.5 m to 1 m, more inter-row shading time will be eliminated than increasing the gap from 2 m to 5 m.

Irradiation per unit land area

As discussed in the last section, when the cost of land is more of a concern than that of a panel, the design criterion becomes the maximization of the irradiation per unit land area. Similarly, when a fixed amount of land is needed to be maximally utilized, the same criterion is applicable. In this section, the same is tested across different layouts and a complete reversal of trend is noticed. In other words, the EWHS layouts outperform the EF layouts in all the cases unlike the last section where the design criterion has been to maximize irradiation per unit panel area.

In Fig. 12(a), we observe that EWHS layout outperforms EF layout for any number of rows and for any tilt angle. The relative reduction in irradiation per land area for EF layouts increases with an increase in tilt. Therefore, for larger value of tilt, the strategy of choosing EWHS layout provides higher advantage. Not only the tilt, but also the size of the array (i.e., number of parallel rows) influences the relative performance of the layouts. From Fig. 12(a), it is evident that the larger the number of rows, the higher is the relative advantage of EWHS layout as compared to its EF counterpart.

One important feature for all the EWHS curves in Fig. 12(a) is a sudden change in slope at 48° tilt angle. The explanation for the same is provided here. The irradiation per land area depends on 3 variables: (i) total irradiation incident on the array, which would follow the same trend as the irradiation per panel area shown in Fig. 12(a), (ii) gap between adjacent rows (shown as G in Fig. 3), and (iii) the projected length of panel on the horizontal plane (shown as L_g in Fig. 3). With increasing tilt angle, G will increase and L_g will decrease. The gaps between rows are introduced to eliminate any shadow cast by one row on another between 10 am and 2 pm throughout the year (i.e., at least four hours of unshaded operation at any given day). According to this criterion, the hut shaped array does not need any gap in between adjacent rows up to 48° tilt angle. However, for higher tilt angles, inter-row gaps are mandatory to avoid shading by the front row for operation within the four-hour window around the noon. Hence, the

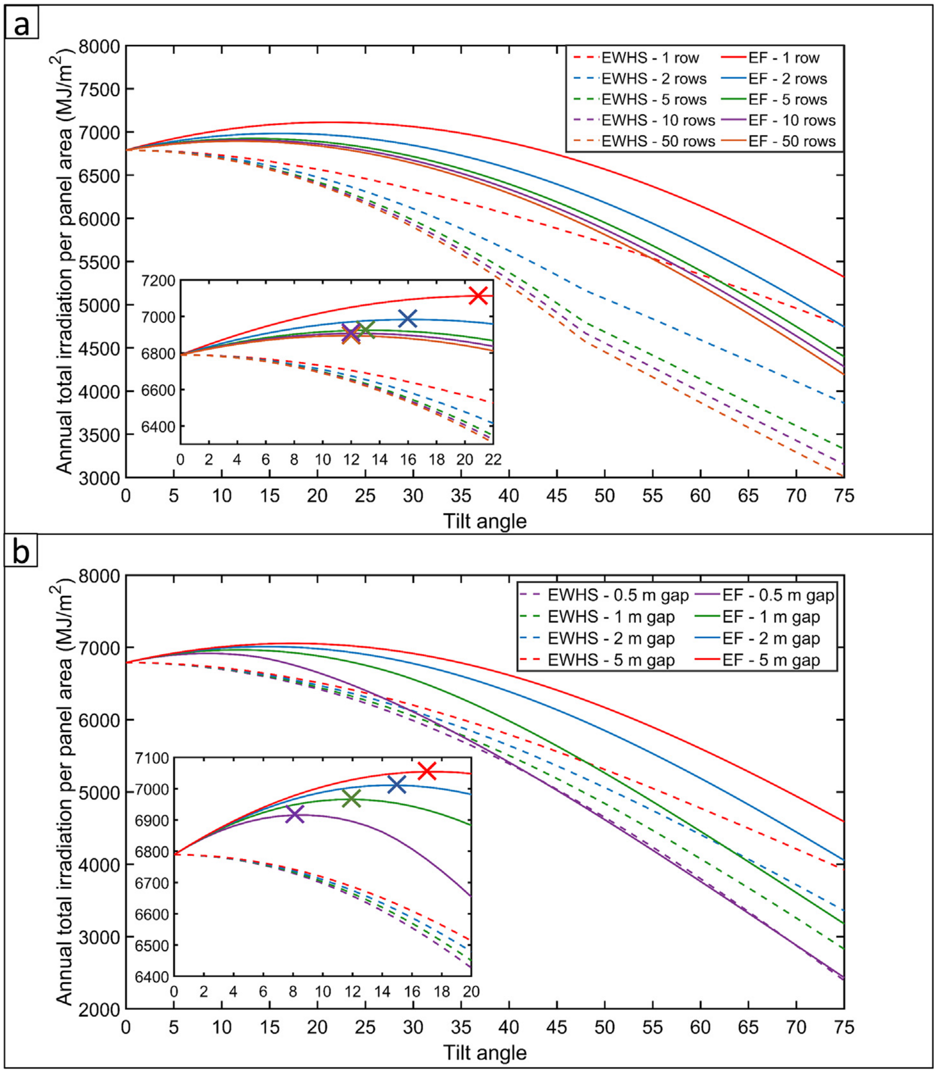

Fig. 11. Annual total irradiation per panel area on south faced array and hut shaped array: (a) Comparison between EF and EWHS arrays with different number of rows without minimum gap between adjacent rows. The inset-figure shows the curves zoomed for 0–22° tilt angle. (b) Comparison between equator facing and EWHS arrays of 50 rows with different gaps between adjacent rows. The inset-figure shows the curves zoomed for 0–20° tilt angle. The points corresponding to the maximum irradiation for equator facing arrays are shown with cross mark on each curve in inset figures. For all EWHS arrays, maximum irradiation is available at 0°, i.e., horizontal position.

introduction of gaps results in EWHS curves changing slope suddenly at that particular tilt angle.

Furthermore, it can be observed from the dashed lines in Fig. 12 (a) that for a small number of rows such as 2 rows, EWHS layout provides better performance per land area for higher value of tilt. The optimal tilts are shown with corresponding cross-marks. However, for greater number of rows, the end effect is negated and the performance degrades beyond a certain critical tilt angle. For example, for a 5-row array, the best performance for EWHS layout is obtained while the tilt angle is 48°. Moreover, for even larger arrays, the change up to 48° tilt is minimal. For a large enough array, the EWHS layout is optimal with zero-tilt, which is equivalent to the horizontal position itself. As horizontal arrangement minimizes the panel area as well, the best strategy to utilize a given land area for maximal irradiation is to cover the area with horizontal panels completely. However, a slight tilt provides the operational advantage of lower soiling due to gravity assisted ease of cleaning.

In Fig. 12(b), same quantities are plotted as in Fig. 12(a). However, a minimum fixed gap of 0.5 m between consecutive rows is introduced in all the layouts to ensure the clearance for maintenance and cleaning. First of all, it is to be observed that the numerical values of irradiation per land area are reduced for all the curves, which happens because of the reduction in panel area due to the introduction of non-productive gap between the rows. Secondly, it is to be noted that the qualitative nature of the dashed lines (corresponding to EWHS arrangements) are less affected in comparison of that of the solid lines (corresponding to EF arrangements). The EF layouts show maxima corresponding to a fixed angle of 12°. The value of this optimal tilt changes with the value of the inter-row spacing, which is not shown here. For the EWHS layouts, the optimal tilt reduces with number of rows, but at much slower rate as compared to Fig. 12(a). The reason behind this is that the inter-row spacing provides an offset in tilt where the shading by the front row becomes significant and alters the annual irradiation substantially.

In Fig. 12(b), sudden changes in slopes of both EF and EWHS curves can be noticed. As discussed before, the gap between adjacent rows is calculated such that there is no inter-row shading between 10 am and 2 pm. Also, in this case, a minimum gap of 0.5 m is kept between rows. Therefore, in the case of EWHS, up to 58° tilt angle, no shadow will be cast on a row by another row between 10 am and 2 pm, i.e., all the shadows will be restricted to the 0.5 m gap. For higher tilt, shadows will be cast on back rows. Therefore above 58° tilt, the gap between rows needs to be higher than 0.5 m to avoid inter-row shading. This

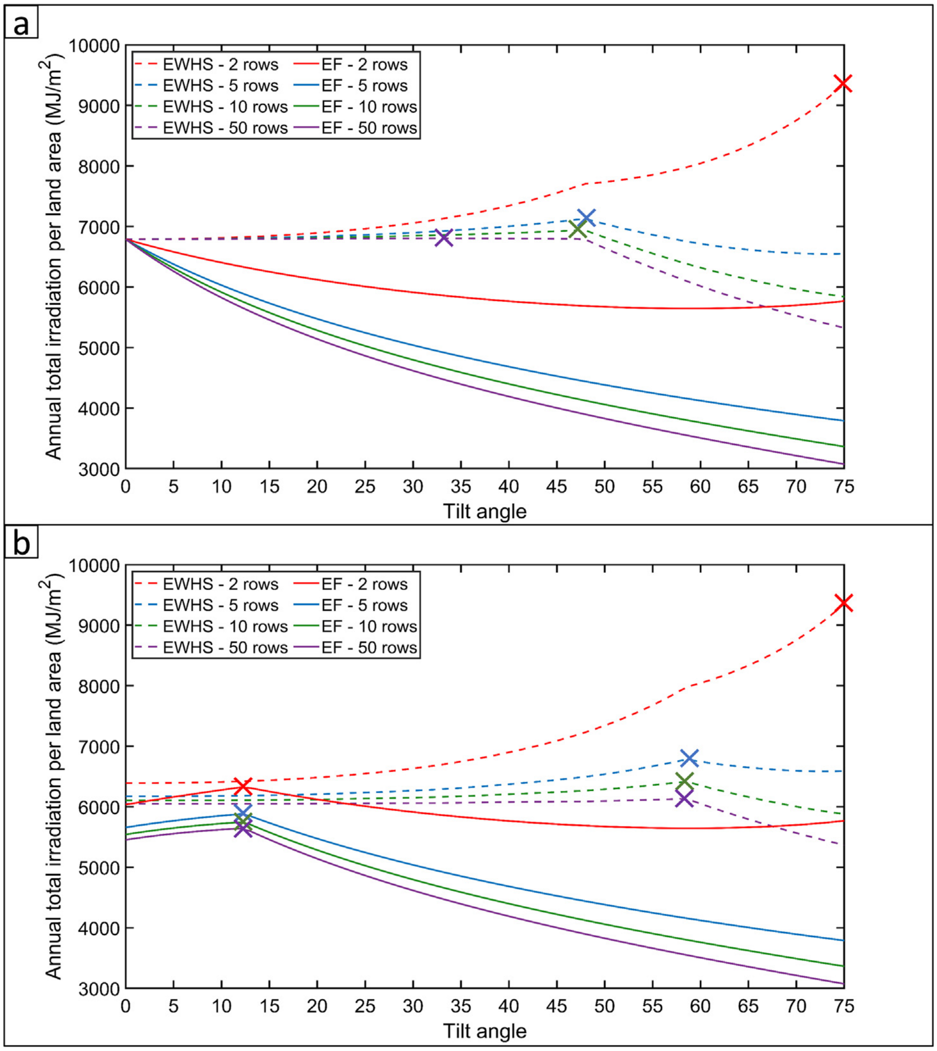

Fig. 12. Annual total irradiation per land area on south faced array and hut shaped array with different number of rows. Cross marks are shown for maximum irradiation on each curve. (a) Array without minimum gap between adjacent rows. For all equator faced arrays (EF) without minimum gap, maximum irradiation is available at 0° tilt, i.e., horizontal position. For EWHS panels, the optimal tilt angle marginally improves irradiation for large enough collector field (50 rows or more) (b) Array with minimum gap of 0.5 m between consecutive rows.

increase in unproductive land area will decrease the irradiation per land area, thus changing the slope of EWHS curve. In the case of equator facing (EF) arrays, there is always a gap between rows to eliminate shading between 10 am and 2 pm, and this gap increases with tilt angle. As a minimum gap of 0.5 m is provided here, which corresponds to the no-shading criterion for 12° tilt, a maximum is observed for each of the EF curves at that particular tilt value. Up to 12° tilt, total irradiation increases as per the same trend as observed in Fig. 11(a). Beyond 12° tilt, the inter-row shading becomes relevant leading to the same trend as observed in Fig. 12(a).

From Fig. 12, it is clear that an EWHS layout is preferred over EF layout when the land area constraint is employed, which is the most significant counter-intuitive observation from the present study, hence finds mention in the title. Without any inter-row spacing, the horizontal arrangement provides the highest annual irradiation while spacing or a slight tilt for EWHS rows can be chosen for gravity-aided cleaning and easier maintenance.

Tropical vs. higher latitude

So far, all the discussions are based on the layout comparisons for an arbitrarily chosen location of Kolkata, India, which is located in the tropical region. As the performance of a solar field is critically dependent on the variation of solar altitude angle, it is important to investigate the effect of latitude on the major observations on layout-comparison discussed in Irradiation on a single panel, Irradiation per unit panel area, and Irradiation per unit land area sections. In this section, we discuss EF vs. EWHS comparisons for large collector arrays for Kolkata, India (22.4°N) and Helsinki, Finland (60.1°N) both for per panel area and per land area.

Fig. 13(a) shows the values of the annual irradiation per panel area on EF and EWHS layouts for Kolkata and Helsinki. Both the curves for Kolkata in this figure are the same as that of 50 row array in Fig. 11 (a). From Fig. 13(a), it is clear that the irradiation available at Helsinki is less than that at Kolkata for all tilt angles. The difference in irradiation per panel area is less for higher tilt angles. This can be seen as both sets of dashed and solid curves moving closer with increasing tilt angle. The optimum tilt angles are marked by crosses on all the curves. Similar to Kolkata, EWHS layout at Helsinki shows the maximal point when the panels are horizontal (3106.6 MJ/m^2). The maximum irradiation per panel area for an EF array at Helsinki is available at 34° tilt (3557.9 MJ/m^2), which is 14.5 % higher than that for the best performing EWHS array (i.e., horizontal). The optimal tilt angle for EF layout in Kolkata is 12° (6892.2 MJ/m^2), for which the irradiation is 1.5 % higher than the optimal EWHS array (i.e., horizontal) (= 6788.9 MJ/m^2). Therefore, with increase in latitude from Kolkata to Helsinki, the relative gain in output from EWHS to EF layout improves substantially.

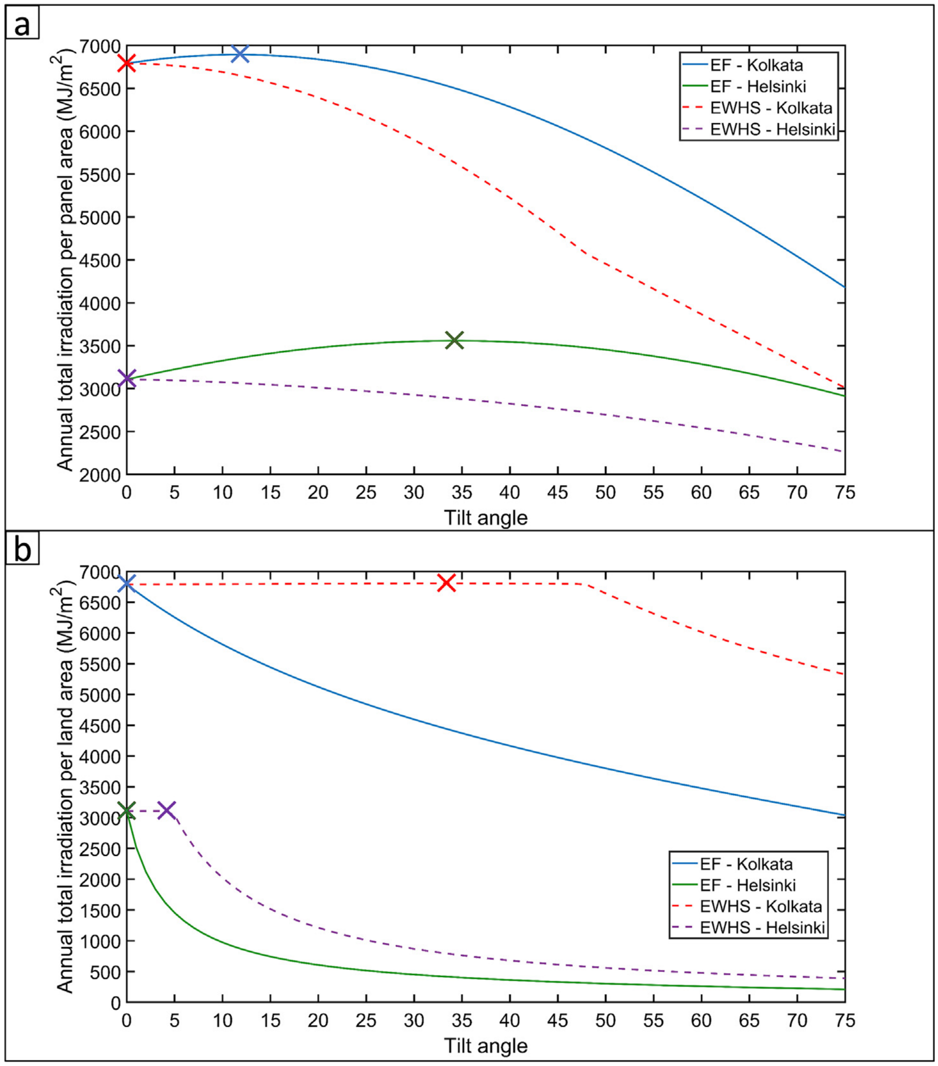

Fig. 13. Comparison between Helsinki (higher latitude) and Kolkata (tropical region): (a) Annual total irradiation per panel area on an EF array and on an EWHS array at Helsinki and Kolkata. (b) Annual total irradiation per land area on an EF array and on an EWHS array at Helsinki and Kolkata. Cross marks indicate maximum irradiation on respective curves.

Since Helsinki is located at higher latitude, the length of the shadow is also higher compared to Kolkata due to lower values of solar altitude angle. The effect of inter-row shading becomes apparent when the comparison metric is chosen to be irradiation per land area. Fig. 13(b) shows the comparison of EF and EWHS layouts at Helsinki and Kolkata in terms of irradiation per land area. Here the EF and EWHS curves for Kolkata are the same as that of 50 row curves in Fig. 12(a). As mentioned before, the spacing between rows is selected in such a way that there is no inter-row shading between 10 am and 2 pm. At Helsinki, due to its higher latitude, the shadow length at 10 am and 2 pm will be longer than that of Kolkata, resulting in longer gaps between rows. The unproductive extra spacing considerably reduces the irradiation per land area for Helsinki. It can be clearly seen from Fig. 13(b) that the irradiation per land area for both EF and EWHS layouts at Helsinki is substantially lower compared to Kolkata.

Similar to Kolkata, the maximal point of EF arrangement at Helsinki is for 0° tilt (3106.6 MJ/m^2), i.e., when the panels are horizontal. For EWHS arrangement, maximum irradiation is available at 4° tilt (3107 MJ/m^2) at Helsinki. Up to 4°, the curve stays minimally upward, beyond which it goes down steeply. The reason for this abrupt alteration in trend is same as discussed in Irradiation per unit land area section. Up to 4° tilt angle, there is no gaps between rows in case of EWHS layout since the inter-row shading between 10 am and 2 pm is absent. For higher tilt than 4°, as the length of shadow increases, it is required to leave gaps between rows which makes the irradiation per land area to drop.

From Fig. 13, two major observations are made. The first is that the change in latitude does not alter the preferred order of the EF and EWHS layouts, i.e., EF is preferred when irradiation per panel area is to be maximized and EWHS is preferred when irradiation per land area is to be maximized. The second observation is that low latitude locations (i.e., Kolkata) provide lower relative advantage for switching to EF from EWHS layout on the basis of irradiation per panel area compared to locations of higher latitude (i.e., Helsinki).

Conclusions

In the current study, total annual solar irradiation is computed for an east-west hut shaped (EWHS) layout of PV panels and it is compared against that for conventional equator facing (EF) layout. The exercise is done for a single panel, array of panels at both tropical and frigid zone locations. The comparison is performed when the constraint is to maximize irradiation per unit panel area as well as per unit land area. Following are the major conclusions of the study.

- The most counter-intuitive fact that came out of these comparisons is that the EWHS array outperforms the conventional EF layouts when land area is the limiting factor rather than the panel area. Hence, when either the available land is limited or the land is the cost-determining component, the designers need to test beyond the conventional EF layout and opt for the optimally tilted EWHS layout to maximize the annual energy yield.
- The second important observation made in the study pertains to the alteration of optimal geometry between a single panel and array of panels in conventional EF layout. The widely accepted norm of choosing optimal tilt equal to the latitude of the location works only for a single panel. For an array, the optimal tilt is very different from the latitude as the inter-row shading effect is included in the irradiation estimate.
- Thirdly, the detailed comparisons in the study confirms that the choice of EF estimate is indeed advantageous when the panel area needs to be minimized, even though the optimal tilt is not the same as the latitude.
- Fourthly, increase in the absolute value of latitude does not alter the conclusion regarding the relative order of preference between EF and EWHS layouts. However, the relative magnitude of advantage changes with latitude. The advantage for an EF array in case of minimization of panel area is higher at higher latitudes compared to the tropical region.
- Finally, when both the land area and panel costs are comparable and optimization in terms of both is targeted, choosing a zero-tilt orientation maximizes irradiation per land area with no increase in panel area. However, the observation is true only when there is no gap in between consecutive rows and there exists no issue of soiling for the panels. The space between consecutive rows is not only necessary for avoiding inter-row shading, but also essential for access for cleaning and maintenance. Moreover, zero-tilt significantly hinders the gravity assisted cleaning of panel soiling thereby compromising long term performance of the overall field. Therefore, the best orientation is to use small tilt angle and inter-row gap for the best performance of a large scale field of fixed angle solar photovoltaic panels.

Thus, the current study provides a detailed analysis of possible array layouts along with the methodology behind it. The key recommendations from the study, as stated above, have direct implication for the designers of such photovoltaic fields. The general applicability of the methodology allows for its implementation for any location and for any relevant design constraints.

Declaration of competing interest

The authors declare no conflict of interest.

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